Pyrroloquinoline Quinone Modulates Mitochondrial Quantity and Function in Mice[1]

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ABSTRACT When pyrroloquinoline quinone (PQQ) is added to an amino acid-based, but otherwise nutritionally complete basal diet, it improves growth-related variables in young mice. We examined PQQ and mitochondrial function based on observations that PQQ deficiency results in elevated plasma glucose concentrations in young mice, and PQQ addition stimulates mitochondrial complex 1 activity in vitro. PQQ-deficient weanling mice had a 20–30% reduction in the relative amount of mitochondria in liver; lower respiratory control ratios, and lower respiratory quotients than PQQ-supplemented mice (2 mg PQQ/kg diet). In mice from dams fed a conventional laboratory diet, but switched at weaning to the basal diet, plasma glucose, Ala, Gly, and Ser concentrations were elevated at 4 wk (PQQ− vs. PQQ+), but not at 8 wk. The relative mitochondrial content (ratio of mtDNA to nuclear DNA) also tended ($P < 0.18$) to be lower (PQQ− vs. PQQ+) at 4 wk, but not at 8 wk. PQQ also counters the mitochondrial complex 1 inhibitor, diphenylene iodonium (DPI). Mice were gavaged with 0, 0.4, or 4 μg PQQ/g body weight (BW) daily for 14 d. At each PQQ level, DPI was injected (i.p.) at 0, 0.4, 0.8, or 1.6 μg DPI/g BW. The PQQ-deficient mice exposed to 0.4 or 4.0 μg DPI/g lost weight and had lower plasma glucose levels than PQQ-supplemented mice ($P < 0.05$). In addition, fibroblasts took up ^{3}H-PQQ added to cell cultures, and cultured hepatocytes maintained mitochondrial PQQ concentrations similar to those observed in vivo. Collectively, these results indicate that dietary PQQ can influence mitochondrial amount and function, particularly in perinatal and weanling mice. J. Nutr. 136: 390–396, 2006.

KEY WORDS: • *pyrroloquinoline quinone* • *mitochondria* • *diphenylene iodonium* • *respiratory control* • *respiratory quotient*

Pyrroloquinoline quinone (PQQ)[3] is a redox cofactor in bacteria and is found in plants and animal tissues in pmol/L to nmol/L concentrations (1–4). PQQ is an aromatic heterocyclic anionic orthoquinone that can reversibly be reduced through a semiquinone intermediate (1–4). PQQ readily reacts with amino acids, alcohols, and nucleophiles to form stable condensation products. In the presence of amino acids, a predominant product is imidazolopyrroloquinoline (IPQ) (5,6). Previous nutritional studies indicate that PQQ can serve as a growth factor in BALB/c mice (7–10); it improves neonatal survival when added to nutritionally complete amino acid–based diets. The response in mice was observed with the addition of as little as 1 nmol PQQ/g of amino acid–based diet. Moreover, in human fibroblast cultures, PQQ and IPQ, a derivative of PQQ, enhance cell growth and proliferation when added to cultures at nmol/L concentrations (11).

Under alkaline conditions, PQQ is also ~100 times more efficient on a molar basis than ascorbic acid, menadione, and typical polyphenolic compounds in assays that assess redox-cycling potential (1,12–15). In bacterial dehydrogenases, PQQ facilitates reductions via mechanisms that are dependent on the transfer of hydride ions (4,6,17). PQQ may also act as an antioxidant (1,18–23).

Given the potential of PQQ as a redox agent, antioxidant, and cell signaling factor [see (1) for review], we hypothesized that these possibilities may affect mitochondrial function. Accordingly, a number of variables important to mitochondrial function were tested including the following: mitochondrial amount, the respiratory control ratio (RCR), and respiratory quotient (RQ).

[1] Supported in part by National Institutes of Health grants DK 35747 and 56031; gifts from M&M Mars, Incorporated (Hackettstown, NJ), Mitsubishi Gas Chemical Company, Incorporated, and the Charitable Leadership Foundation (Clifton Park, NY).

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[3] Abbreviations used: BW, body weight; CF, cystic fibrosis; DPI, diphenylene iodonium; DMSO, dimethyl sulfoxide; GDH, glucose dehydrogenase; IPQ, imidazolopyrroloquinoline; NBT, nitro blue tetrazolium; ND-5, nicotinamide adenine dinucleotide dehydrogenase-5; PQQ, pyrroloquinoline quinone; PQQ+, PQQ supplemented; PQQ−, PQQ deficient; qRT-PCR, quantitative RT-PCR; RCR, respiratory control ratio; RQ, respiratory quotient.

MATERIALS AND METHODS

Reagents. Chemicals and reagents used in diets and assays were obtained from Fisher Chemicals and Sigma-Aldrich and were of the

Manuscript received 16 July 2005. Initial review completed 16 August 2005. Revision accepted 21 November 2005.

highest purity available. Amino acids (for diet preparations) were obtained from Ajinomoto. PCR primers were purchased from Invitrogen and reagents for qRT-PCR were purchased from Applied Biosystems. Cell culture supplies were obtained from Sigma-Aldrich, Becton Dickinson, and Calbiochem. Rats (Sprague-Dawley) and mice (BALB/c and C57BL/6J) were purchased from Charles Rivers.

Animal care, diets and nutritional protocols. The mice were housed and maintained in an AALAC approved facility with approval from the University of California, Davis Institutional Animal Care and Use Committee. Mice and rats had free access to food and water. The composition of the diet is given in **Table 1**. PQQ concentrations in the basal amino acid diet were determined to be <5 fmol/g diet using a glucose dehydrogenase (GDH) enzyme assay for PQQ quantitation (24,25); see below. In Expts. 1 and 2, BALB/c mice were used. PQQ was added to diets at 2 mg/kg (~6 nmol/g) diet. In typical protocols, virgin females were adapted to the basal (devoid of PQQ) diet for ≥2 wk. Half of the females were then switched to a diet supplemented with 2 mg PQQ/kg diet, and the other half were fed the basal diet. Both groups of females (F_o generation) were then bred; the resulting pups (F_1 generation) were weaned at 28 d of age and fed the same diet as their dams until used in selected assays. In Expt. 1, mitochondrial morphometric, RCR, and RQ measurements were made at d 7 after weaning or 35 d postpartum. In Expt. 2, the interaction between PQQ and diphenylene iodonium (DPI) was determined. Other details regarding husbandry were described (8). In Expt. 3, C57BL/6J mice were chosen because of their different genetic background and to extend the findings of Expts. 1 and 2. In contrast to Expts. 1 and 2, the C57BL/6J mice were derived from dams fed a conventional diet and switched to PQQ-deficient or -supplemented diets at weaning or 28 d postpartum.

Expt. 1: PQQ and mitochondria in vivo. Mitochondrial morphometric and respiration measurements were performed in Expt. 1. For morphology, liver samples were cut into small (<2 mm × 2 mm) pieces and initially fixed in 2.5% glutaraldehyde buffered with 0.1 mol/L cacodylic acid buffer (pH 7.4). The samples were fixed further in 1% osmium tetraoxide. After the imbedding of the samples in a Bojax plastic mixture (epon/araldite), they were sectioned at 600–900 Å, and stained with 2% lead citrate/uranyl acetate. The cells were examined using a transmission electron microscope (EM10AZeiss, LEO Electron Microscopy). Photomicrographs of 10 individual cells from each of 6 mice/group were examined at a magnification of 2000X. An NIH image program (26) was used to calculate the cell, nuclear, and mitochondrial size, and the number of mitochondria per cell. In anticipation that mitochondrial cellular quantity might be influenced by differences in body weight (BW), mice were weight-matched. The mice were selected so that BW were similar in both groups (i.e., 10–12.7 g).

Estimates for mitochondrial respiration were carried out using a Clark-type oxygen electrode and a mitochondrial respiration chamber (Yellow Springs Instrument) as described by Bobyleva-Guarriero et al. (27,28). Male BALB/c F_1 offspring at 35 d postpartum were used.

For mitochondrial isolations, whole livers were removed and minced into ~1.5-mm pieces, and gently homogenized (10% wt:vol) into 0.25 mol/L sucrose containing 3 mmol/L HEPES buffer (pH 7.4) and 5 mmol/L EGTA using 5 strokes of a Potter-Elvehjem tissue homogenizer rotating at 1200 rpm. These homogenates were centrifuged at 900 × *g* for 10 min. The supernatant fraction was collected and recentrifuged at 14,500 × *g* for 10 min. The resulting pellets were washed twice by resuspension in 20 mL of isolation medium (EGTA omitted) followed by centrifugation (9000 × *g* for 10 min).

For substrates, α-ketoglutarate (10 mmol/L) or succinate (50 μmol/L) was used to assess complex I or complex II activities, respectively. To determine the RCR, the contents of the incubation chamber were continually stirred and a basal rate of O_2 consumption was measured. Substrates, ADP, and inhibitors were added in volumes <40 μL. Two additions of ADP were made. State 3 respirations (ADP and substrate present) were determined from O_2 uptake rates during the second ADP addition, and state 4 respiration (only substrate present) was determined after the decline in rate when the second addition of ADP was metabolized. The RCR was calculated as the state 3 rate compared with the state 4 rate. For all assays, the Clark oxygen electrodes were calibrated with phenylhydrazine as described by Misra and Fridovich (29); the results were expressed per mg protein (30).

Respiratory quotient. Mice from Expt. 1 were also used for RQ measurements. They were first weighed and placed into individual Plexiglas metabolic chambers submerged in a water bath. The temperature was initially set at 28°C and monitored throughout with an YSI (Yellow Springs) thermister. Oxygen consumption and CO_2 were measured in an open system using an Ametek S3-A oxygen analyzer (Advanced Micro Instruments). The S3-A analyzer detects oxygen differences with a precision of ±0.01%. The dried airflow rate was set at 200 mL/min and the chamber size was ~200 mL. The time for washout of the system was ~5 min at the 200 mL/min flow rate. Results were corrected to standard temperature and pressure, and oxygen volume was calculated (31,32) using a separate analyzer-recorder system.

For RQ estimation, pups were allowed to acclimate to the chamber (~30 min at thermoneutrality, i.e., 28°C). Resting rates were recorded over a 30-min period such that for every pup, 3–4 values were obtained for each recording period (5 min). Next, the chamber temperature was lowered to 16°C, at the rate of 1°C every 5 min. Mice were then held at 16°C and additional measurements were made over a 4- to 5-h period.

Expt. 2: DPI. Male BALB/c offspring from PQQ-deficient dams (at 35 postpartum) were gavaged daily (10 μL/g BW) with PQQ for 10 d in amounts corresponding to 0, 0.4, or 4.0 μg PQQ/g mouse, i.e., from 0 to ~12 nmol PQQ/g mouse. PQQ was dissolved in PBS (pH 7.0). Immediately after PQQ administration, the mice were given an i.p. injection of DPI at doses of 0, 0.4, 0.8, or 1.6 μg DPI/g mouse, i.e., from 0 to ~5 nmol DPI/g mouse for each level of PQQ. On d 45 postpartum, blood samples were collected 3 h after the PQQ gavage and DPI injections. Mice were food deprived during this time period. The mice were weighed before killing, necropsied, and whole-blood samples were analyzed for glucose (Sigma Chemical) based on glucose oxidase/peroxidase O-dianisidine oxidation.

PQQ, DPI, and mitochondria in vitro. Liver mitochondria isolated from rats fed a standard diet were used to assess in vitro the interaction between PQQ and DPI and to estimate mitochondrial PQQ content. Mitochondria were isolated as described above. Rat liver rather than mouse liver was chosen to obtain sufficient quantities of mitochondria to facilitate isolation of PQQ and estimation of mitochondrial diaphorase.

To isolate putative PQQ in mitochondria, mitochondrial pellets (~1-g aliquots) were collected, resuspended in H_2O (30% wt:v), and pulse-sonicated for 4 min in a vessel surrounded by crushed ice. The

TABLE 1

Diet Composition

Diet ingredient	*g/100 g*
Amino acid mix[1]	18.0
Sucrose	31.5
Cornstarch	27.5
Corn oil	10.0
Mineral mix[2]	10.0
Cellulose	2.5
Vitamin mix[3]	0.5

1 Amino acids (g/kg diet): L-arginine·HCl, 12; L-proline, 10; L-serine, 12; L-histidine, 4; L-isoleucine, 8; L-leucine, 12; L-lysine·HCl, 14; L-methionine, 6; L-phenylalanine, 5; L-tyrosine, 5; L-threonine, 8; L-tryptophan, 2; L-valine, 8; L-alanine, 12; L-asparagine, 11.5; L-cystine, 4; L-glutamate, 12.5; glycine, 12.

2 Minerals (g/kg diet): sodium carbonate, 3.0; calcium phosphate, 28.0; potassium phosphate dibasic, 9.0; sodium chloride, 8.8; sodium bicarbonate, 10.0; magnesium sulfate·H_2O, 1.85; manganese sulfate·H_2O, 0.650; ferric citrate, 0.5; zinc carbonate, 0.1; copper sulfate·$5H_2O$, 0.05; boric acid, 0.005; sodium molybdate·$2H_2O$, 0.005; sodium fluoride, 0.005; nickel(II) chloride·$6H_2O$, 0.005; potassium iodide, 0.001; cobalt (III) sulfate·$7H_2O$, 0.001; chromium (III) chloride·$6H_2O$, 0.001; sodium selenite (as mg/kg), 0.250; and cornstarch to 100.0 g.

3 Vitamins (mg/kg diet): choline·HCl, 2000; thiamine·HCl, 100; niacin, 50; riboflavin, 15; calcium pantothenate, 30; vitamin B-12, 0.2; pyridoxine·HCl, 30; biotin, 3; folic acid, 20; myoinositol. 100; ascorbic acid, 250; retinyl acetate 1.8; cholecalciferol 0.025; *dl*-α-tocopherol acetate, 40; phylloquinone, 1.0; and glucose to a total of 5.0 g.

suspension was transferred to dialysis tubing with pore size corresponding to a MW of 10,000 and dialyzed against H_2O at 4°C for 24 h. The dialysate was collected and quickly lyophilized. Lyophilized material was resuspended in H_2O. The samples were then chromatographed as described previously by Fluckiger et al. (12,13) and Paz et al. (14,15). The putative PQQ in mitochondria was defined by co-migration with authentic PQQ and its ability to carry out redox cycling and serve as a cofactor in the GDH assay for PQQ (25,26). In addition, the material co-migrating with PQQ was further characterized on the basis of the patterns of inhibition observed when indium sulfate, lead acetate, or phenazine methosulfate was added to assays (12,15). Diaphorase activity was chosen because PQQ was shown to serve as an electron accepter when a mitochondrial complex I substrate is used (33). A combination of PQQ and/or DPI, plus either the complex I substrate, α-ketoglutarate, or the complex II substrate, succinate, was used for the assay. The assay mixture also contained 10 mmol/L $MgCl_2$ plus 100 μL nitro blue tetrazolium (NBT) (5 g/L saturated solution). The assay mixtures were adjusted to 500 μL with PBS (pH 7.4). Mitochondrial protein (1 mg) was added to start the reactions, followed by shaking at 37°C for 45 min. The samples were then centrifuged (15,000 $\times$ g, 30 min at 4°C). The supernatant fraction was discarded and 250 μL dimethyl sulfoxide (DMSO) was added, followed by vigorous mixing on a vortex and centrifugation (1000 $\times$ g) to separate formazan (oxidized NBT) from the mitochondrial pellet. For assays, 200-μL aliquots were transferred to multiple well plates for estimation of formazan (12–15).

Plasma glucose, lactic acid, amino acids, and mitochondrial DNA quantitation. In Expt. 3 (C57BL/6J mice), plasma glucose and lactic acidwere determined using commercial assay kits (QuantiChrom™ glucose assay kit, [DIGL-200], Sigma Chemical Co.; Lactic acid assay kit [K-DLATE], Megazyme International Ireland Ltd.); plasma amino acid concentrations were determined using a model 7300 Beckman Amino Acid Analyzer (0.4 cm $\times$ 10 cm column packed with spherical cation exchange resin; Beckman Instruments). Before analysis, plasma was mixed with an equal volume of 0.28 mol/L 5-sulfosalicylic acid. The resulting precipitate was removed by centrifugation at 16,000 $\times$ g. Lithium hydroxide was added to an aliquot of the supernatant to adjust the pH to 2.2 and the equivalent of 20 μL of plasma was injected onto the column of the analyzer. Norleucine was used as an internal standard [see (34–36) for additional details].

The relative amounts of liver mitochondria in C57BL/6J mice (Expt. 3) were determined using RT-PCR methods described by Wong and Cortopassi (37). The targeted genes were nuclear cystic fibrosis (CF) and mitochondrial nicotinamide adenine dinucleotide dehydrogenase-5 (ND-5). For nuclear DNA quantification, 10 ng DNA was used as a template. Mouse specific primers were selected using Primer Express® Software (Applied Biosystems). Primers for CF were: forward 5′-TGT TGT GAA GAC GAG CTG ATG TAA AG-3′; reverse 5′-TGC ATT AAA AGA GAG CAT GTG TTG-3′. For mitochondrial DNA quantification, 0.1 ng DNA was used as a template and primers for ND-5 were: forward 5′-TGG ATG ATG GTA CGG ACG AA-3′; reverse 5′-TGC GGT TAT AGA GGA TTG CTT GT-3′. Qualitative RT-PCR (qRT-PCR) was performed using a ABI 7900HT real-time thermocycler coupled with SYBR Green technology (Applied Biosystems) and the following cycling parameters: stage 1, 50°C for 2 min; stage 2, 95°C for 10 min; stage 3, 40 cycles for 95°C for 15 s; 60°C for 1 min; and stage 4, 95°C for 15 s; 60°C for 15 s; 95°C for 15 s. The linearity of the dissociation curve was analyzed using ABI 7900HT software, and the mean cycle time of the linear part of the curve was designated Ct. Each sample was analyzed in duplicate. Relative mitochondrial copy number to nuclear copy number was assessed by a comparative Ct method, using the following equation:

$$\Delta Ct_{\text{mitochondria/nuclear}} = Ct_{\text{mitochondria Ct}} - Ct_{\text{nuclear Ct}}.$$

The fold changes relative to control rats were calculated using the following equation: $2(-\Delta\Delta Ct\text{mitochondria/nuclear})$, where $\Delta\Delta Ct_{\text{mitochondria/nuclear}}$ = mean $\Delta Ct_{\text{mitochondria/nuclear}}$ of the control animals $-$ $\Delta Ct_{\text{mitochondria/nuclear}}$ of each animal from different dietary groups. Values represent mean fold change $\pm$ SD.

PQQ Estimation. A GDH-based assay system was used for PQQ quantitation (24,25). Recoveries of spiked samples (plasma and liver extracts) were good (e.g., >85%). The extraction protocol for PQQ was the same as that described in Steinberg et al. (8).

Cellular Uptake of PQQ. Human fibroblast cells (GM05565, Coriell Cell Repositories) were grown to confluence in DMEM containing 10% fetal bovine serum and penicillin/streptomycin in 175 cm^2 culture plates. The medium was aspirated and the cells were rinsed with serum-free DMEM medium containing *mito+serum extender*™ (Becton Dickinson) for 10 min. This medium was also aspirated and serum-free DMEM medium readded. The cells (15 $\times$ 10^6/plate) were then inoculated with 45 μCi, (16.6 kBq) of ^{3}H-PQQ adjusted to give 20 nmol/L. The cells were incubated in triplicate for 0, 1, 2, 4, 8, 16, and 24 h. At the end of the incubation period, the medium was aspirated and saved for liquid scintillation counting. Labeled ^{3}H-PQQ was prepared by DuPont-New England Nuclear using the Wilzbach method (38). Before use, exchangeable tritium was removed from the ^{3}H-PQQ product by chromatography on columns of A-25 Sephadex followed by absorption, elution (5 times) from C18-Octadecyl columns, and characterization [Amersham Biosciences; see (5,6,25,26)].

To prepare cellular fractions, the cells were trypsinized and centrifuged at 1000 $\times$ g for 5 min. Nonspecific binding of PQQ was decreased by first aspirating the supernatant, and then suspending the cells in ice-cold isolation buffer [210 mmol/L mannitol, 70 mmol/L sucrose, 1 mmol/L EGTA, 0.5% bovine serum albumin (fatty acid free), 5 mmol/L HEPES at pH 7.2] containing 10 μmol/L nonradioactive PQQ. The cells were incubated for 30 min and again recovered by centrifugation at 1000 $\times$ g for 5 min. Next, digitonin (10% wt:v in DMSO) was added to a final concentration of 0.1 g/L or until >95% cell lysis was achieved. Trypan blue exclusion determined cell permeabilization. The permeabilized cells were suspended in isolation buffer and disrupted with 20 passes of a chilled Dounce Homogenizer with a tight-fitting pestle. To obtain a nuclear enriched fraction, the homogenate was centrifuged at 3000 $\times$ g for 5 min. The supernatant fraction was centrifuged at 10,000 $\times$ g for 20 min to obtain a fraction enriched in mitochondria. This fraction was sonicated and saved for liquid scintillation counting. For a microsomal enriched fraction, the resulting supernatant fraction after isolation of the mitochondria was recentrifuged at 100,000 $\times$ g for 60 min. The pellet was suspended with agitation, sonicated, and saved for determination of radioactivity by scintillation counting.

In addition, normal mouse liver and mouse Hepa 1-6 cells were used as sources of tissue for PQQ quantitation. The Hepa 1-6 cells were cultured (at 37°C with 5–10% CO_2, 24 h) in 90% DMEM (4.5 g/L glucose) plus 10% fetal calf serum to which 0, 15, or 30 μmol/L PQQ was added. The cells were harvested, washed twice with PBS and homogenized in isolation buffer (10 mmol/L Hepes pH 7.8, 0.2 mmol/L EGTA, and 0.25 mol/L sucrose). The cell homogenate was centrifuged at 1000 $\times$ g for 10 min at 4°C. The resulting supernatant was then centrifuged at 12,000 $\times$ g for 15 min at 4°C for separation of a mitochondrial enriched pellet and cytosol (supernatant) fractions. Values for PQQ obtained from the GDH assay were normalized per mg of protein (29).

Statistical analysis. The Statview 5.0 statistical analysis program (SAS Institute) was used to analyze the results. Results were analyzed using a t test or ANOVA using a Bonferroni post hoc test. Results are presented as means $\pm$ SEM or SD. Differences were considered significant at $P < 0.05$.

RESULTS

Mitochondrial function. The amount of mitochondria in liver from PQQ-deprived mice was less than that from PQQ-supplemented mice based on cross-sectional area estimates (**Fig 1**, **Table 2**). In addition to the reduction in mitochondria, another striking feature was the difficulty in obtaining values for RCR from the mitochondrial preparations obtained from PQQ-deprived mice compared with supplemented mice (Table 2). Because mitochondrial preparations from fetal and neonatal mice have more permeable inner membranes, relatively low values for RCR were expected (39–44). However, only RCR values >1.8 were used for data summaries. Over half of the mitochondrial preparations from PQQ-deficient mice exhibited

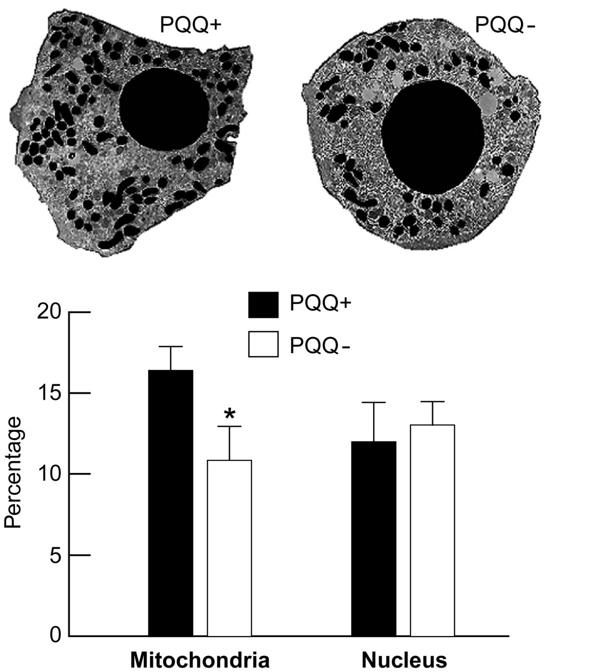

FIGURE 1 PQQ dietary status influences mitochondrial content in BALB/c mice (Expt. 1). The area occupied by mitochondria in liver cells from PQQ-deprived mice was reduced compared with PQQ-supplemented mice ($P < 0.05$). The small darkened areas correspond to mitochondrial cross-sectional areas; selected associated variables are summarized in Table 2. Values are means ± SEM, $n = 6$. *Different from PQQ+, $P < 0.05$.

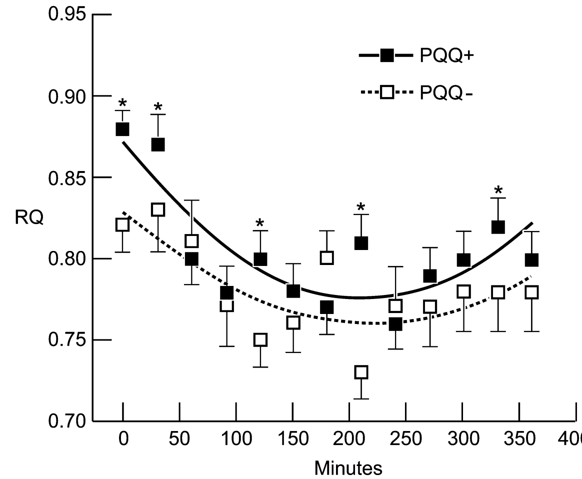

FIGURE 2 RQ values for BALB/c mice with and without PQQ administration in response to a temperature change from 28 to 16°C. The temperature was lowered 1°C every 5 min. Measurements were made on all of the PQQ-supplemented mice over the 6-h period ($n = 9$); however, only 6 PQQ-deficient mice tolerated exposure to 16°C beyond 200 min in the group. The overall change in RQ was significantly influenced by temperature and diet (ANOVA using a Bonferroni post hoc test, $P < 0.05$). *Different from PQQ+ at that time point, $P < 0.05$.

little or no respiratory control compared with PQQ-supplemented mice. Although the values presented in Table 2 for RCR only approach significance ($P = >0.15$), they do not include values from the mitochondrial preparations from PQQ-deficient mice with values of RCR <1.8.

PQQ and RQ. The RQ values for PQQ-deprived mice (Expt. 1) were reduced compared with PQQ-supplemented mice (**Fig. 2**). The values remained consistently lower in response to decreasing the temperature to 16°C. In addition, only two-thirds of the PQQ-deficient mice tolerated the decrease in temperature, whereas all of the PQQ-supplemented mice tolerated the procedure without displaying listlessness or labored breathing. For PQQ-deficient mice, these signs occurred within 1 h of exposure to 16°C.

DPI and PQQ. Results for the interaction between DPI and PQQ (Expt. 2) are given in **Figure 3** *A* and *B*. When DPI was administered (i.p.) to PQQ-derived mice, there was a decrease in the rate of growth of mice that was reversed by oral PQQ supplementation. However, an intake of an equivalent amount of PQQ was adequate to improve growth and circulating glucose levels. Mice supplemented with PQQ had normal whole-blood glucose levels and grew at rates equal to PQQ-supplemented mice without DPI exposure. Different routes of administration of PQQ (oral gavage) and DPI (i.p.) were chosen to avoid direct chemical interaction between the 2 compounds.

When PQQ was added to rat liver mitochondrial preparations with or without DPI additions, there was an increase in diaphorase activity (Fig. 3 C). The increase in diaphorase activity occurred only when α-ketoglutarate, a complex I mitochondrial substrate, was used to initiate the reaction; it did not occur when succinate, a complex II substrate, was used (results not shown).

PQQ, mtDNA, plasma glucose, lactic acid, and amino acids. There was an elevation in plasma glucose, alanine threonine, serine, glycine, tyrosine, methionine, and ornithine in plasma from PQQ-deficient C-57 mice compared with PQQ-supplemented mice at 4 wk, but not at 8 wk postweaning (Expt. 3, **Fig 4** *A* and *B*). The decrease in liver mitochondrial DNA to nuclear DNA, although not significant ($P < 0.18$), was considered consistent with this observation given that the amino acids particularly dependent on mitochondria for oxidation (e.g., alanine, serine and glycine) were elevated (see Discussion). There were no differences in any of the variables at 8 wk.

Isolation and characterization of PQQ in mitochondria. Mitochondrial preparations contained substances that co-migrated with PQQ standards and exhibited high redox-cycling capacity. Two peaks with redox cycling activity were observed at 7–9 and 31–32 min (elution position of authentic PQQ) (**Fig. 5**). PQQ and the putative PQQ fraction also responded in an identical fashion when redox cycling (reduction of nitroblue tetrazolium to formazan) was initiated by PQQ in the presence of glycine. The major fraction co-migrating as authentic PQQ (31–32 min) catalyzed robust redox cycling at pH 10 (12–15). In the presence of redox cycling inhibitors [see (12,13)], indium chloride, lead acetate, or phenalazine methosulfate (10 μmol/L to assays), inhibition in redox cycling was 70–75, 55–60, or 40–60%, respectively, for PQQ and the mitochondrial fraction co-migrating as PQQ.

TABLE 2

Mitochondrial content and the respiratory control ratio for liver from PQQ-deficient and -supplemented BALB/c mice[1]

Item	PQQ+	PQQ −
RCR	2.44 ± 0.12	2.16 ± 0.24
Functional mitochondrial preparations, %	78	29*
Mitochondria, *n/cell*	91.0 ± 6.6	56.8 ± 7.8*
Size of individual mitochondria, μm^2	0.823 ± 0.070	0.808 ± 0.081

[1] Values are means ± SEM or %, $n = 6$. *Different from PQQ+, $P < 0.01$.

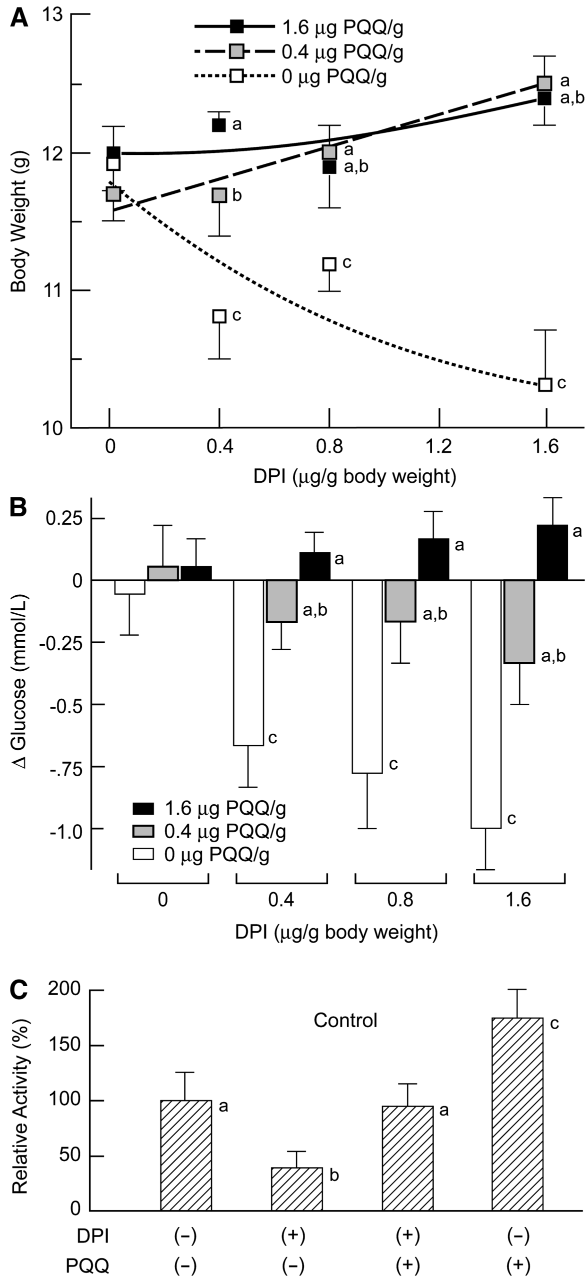

FIGURE 3 Changes in body weights (*A*) and serum glucose (*B*) after 10 d of DPI treatment in PQQ+ and PQQ− BALB/c mice (Expt. 2) demonstrating the interaction between DPI and PQQ. The glucose concentration of the designated food-deprived control group (0.4 μg PQQ/g mouse) was 4.3 mmol/L (78 ± 6 mg/dL) glucose. The mice weighed 11.6 g ± 1.1 g immediately before DPI administration. Values are means ± SEM, $n = 4$. Means at a DPI dose without a common letter differ, $P < 0.05$. PQQ was also effective in reversing DPI inhibition of diaphorase activity (*C*) when added to mitochondrial preparations in assays in vitro. α-Ketoglutarate was used as the complex I substrate. Values are mean % ± SD, $n = 3$. Means without a common letter differ, $P < 0.05$. Using succinate to estimate complex II activity (as an additional control), values for diaphorase were similar to blanks and were not affected by DPI or PQQ additions (results not shown).

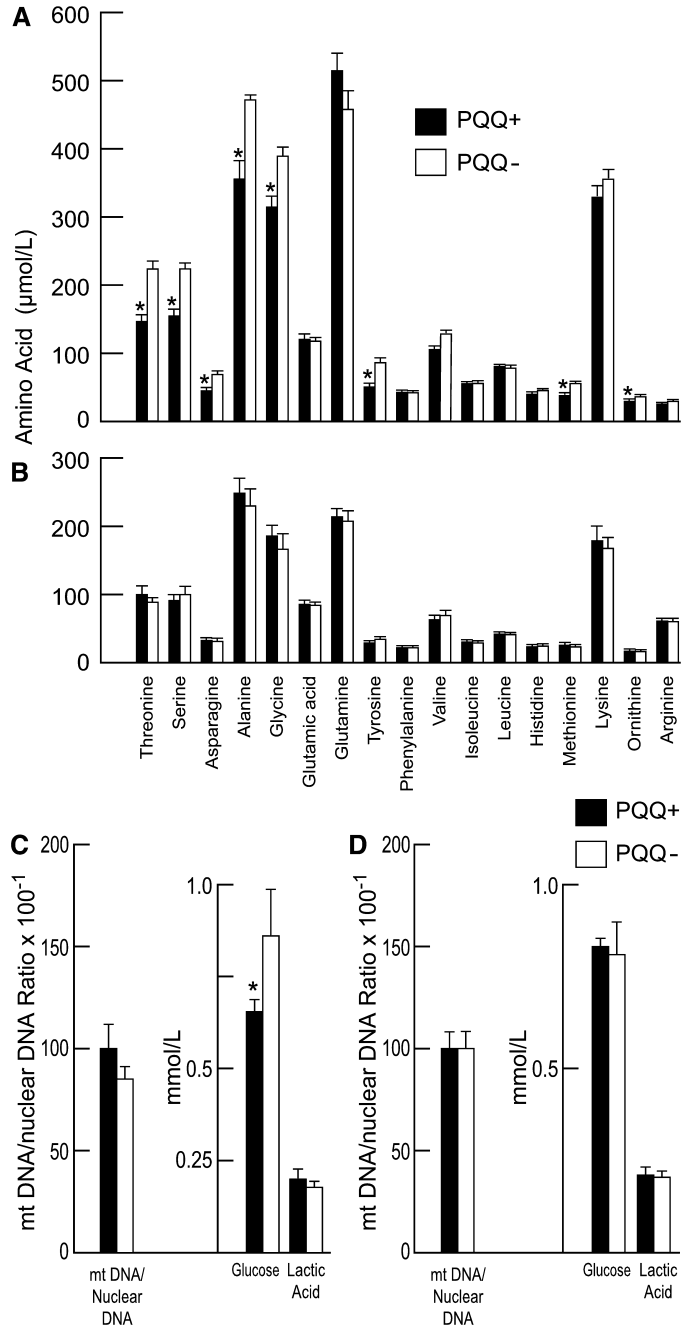

FIGURE 4 Changes in plasma glucose, lactic acid, selected amino acids and the liver mtDNA to nuclear DNA ratio in BALB/c mice fed diets deficient or supplemented with PQQ. Values for plasma amino acids from mice at 4 or 8 wk postweaning are shown in *panels A* and *B*, respectively. Values for glucose, lactic acid, and the mtDNA/nuclear DNA ratio at 4 and 8 wk postweaning are shown in *panels C* and *D*, respectively. Values are means ± SEM, n = 4–6. *Different from PQQ+ for the amino acids indicated and glucose at wk 4, $P < 0.05$.

Cellular uptake of PQQ. It could also be demonstrated that cells efficiently take up ^{3}H-PQQ added to cultures of human fibroblasts. The overall uptake of PQQ into cells was time dependent (**Fig. 6**). The total incorporation of ^{3}H-PQQ into the cells at 24 h was 0.37 ± 0.01% of the total radiolabel of ^{3}H-PQQ added to cultures. PQQ was progressively incorporated into various cell fractions, except for the pellet corresponding to microsomes. At 24 h, cytosol contained 73%, the cell membrane plus nuclear pellet contained 11%, and the mitochondria contained 16% of the total ^{3}H-PQQ taken up by fibroblasts. Moreover, in a separate experiment using mouse Hepa 1-6 cells, PQQ could also be detected in the mitochondria and cytosolic fractions. After 24 h in culture, values for PQQ in whole-cell extracts were 3.3 ± 0.6, 3.5 ± 0.4, 3.9 ± 1.1, and 4.2 ± 0.8 pmol/mg protein at additions of 0, 5, 15, or

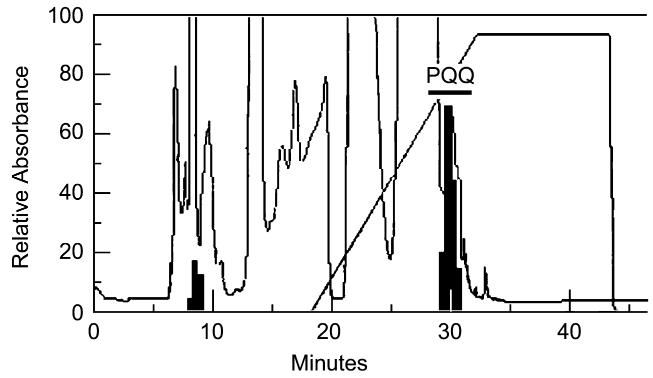

FIGURE 5 Compounds capable of redox cycling were observed in mitochondrial extracts; they co-migrated with authentic PQQ (defined by the labeled bar) after HPLC separation.

30 nmol PQQ/L of medium, respectively. Values for PQQ in the corresponding mitochondrial fractions from the same cells were 4.1 ± 0.6, 4.8 ± 0.4, 5.1 ± 0.7, and 4.1 ± 0.8 pmol PQQ/mg protein, respectively.

DISCUSSION

Each of the observations reported provides evidence or validates in part that PQQ or a related derivative (e.g., IPQ,) is important to mitochondrial function. Moreover, it is of interest that it is the lack of PQQ that led to an apparent decrease in mitochondrial content or perturbation in function. Further, relatively low PQQ concentrations (nmol/g of diet) and low tissue concentrations, pmol/L to nmol/L, were sufficient to reverse the response.

For some of the observations, it was important to eliminate the rate of growth as a factor (40,41). Accordingly, for morphological assessments of mitochondria, mice from PQQ-deficient and -supplemented groups were matched for weight, yet a reduction in mitochondrial content persisted. In addition to the decrease in mitochondrial number, there was also difficulty in assessing the RCR using mitochondrial isolates from PQQ-deprived mice compared with PQQ-supplemented mice. Fewer than half of the mitochondrial preparations from PQQ-deficient mice could be used for RCR measurements. In some respects, PQQ-deprived mice had mitochondria that resembled those from fetal or neonatal mice [e.g., low values for state 3 and 4 oxidation and less efficient respiratory coupling than adult mice (40–44)].

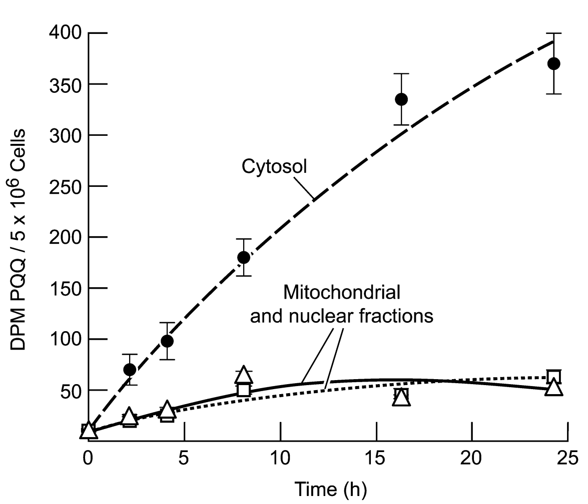

FIGURE 6 ^{3}H-PQQ incorporated into human fibroblasts. Of the total radiolabel, 16 and 11% was present in the mitochondrial and nuclear fractions, respectively, at 24 h; the remainder was associated with the cytosol. Values are means ± SEM, $n = 3$ separate cultures.

We previously reported that as mice age, or when weaned from dams fed a laboratory diet, changes in PQQ status initiated after weaning had less effect on growth-related variables (1,7,8). For example, BALB/c mice reared from dams fed amino acid-based diets without PQQ supplementation eventually catch up with respect to weight gain as they approach sexual maturation (7,8). Consequently, it was viewed as important that changes in mitochondrial amount and differences in plasma glucose and amino acids were more obvious in weanling and 4 wk-old mice than in 8 wk-old mice. The elevation in glucose was considered in part consistent with diminished mitochondrial function (45,46). As Oda et al. (45) and Xue et al. (46) observed, depending on the species, the metabolism of L-serine, L-alanine, and glycine occurs mainly in mitochondria and/or peroxisomes. Serine:pyruvate/alanine:glyoxylate aminotransferase and the mitochondrial glycine cleavage system, which is localized in peroxisomes and mitochondria in rats and mice, are quantitatively more important to serine metabolism than flux through serine dehydratase (45).

In addition, the lower values for RQ in PQQ-deficient mice compared with PQQ-supplemented mice were also taken as a sign of compromised mitochondrial function and inefficient carbohydrate utilization (47,48). The RQ of weaning mice is normally between 0.9 and 1.0. Although the response to lowering the ambient temperature was similar in both PQQ+ and PQQ− mice (e.g., a decrease in RQ followed by adaptation), only two-thirds of the PQQ-deprived mice tolerated the protocol.

PQQ-deprived mice were also very sensitive to DPI, a potent antiglycemic agent and complex I mitochondrial inhibitor (49–51). DPI inhibits the mitochondrial NADH-ubiquinone oxidoreductase (complex I) on the substrate side of Fe-S clusters associated with complex I and is thought to react irreversibly with FMN (50). Because many quinones are good electron acceptors in assays for diaphorase and complex 1 activity (33), a direct connection between complex I activity and PQQ is tenuous; nevertheless, it was viewed as important that PQQ administered orally to mice counters the effects of DPI administered i.p. or on complex I activity in vitro.

As a final point, putative PQQ is present in mitochondrial fractions based on chromatographic behavior, response to inhibitors in PQQ-based redox cycling assays, and its detection in highly sensitive and specific GDH assays. Based on estimates from GDH assays, the amount of free-PQQ in liver mitochondria was ~4 pmol/mg of mitochondrial protein when cells were grown under standard culture conditions. Such values are similar to those reported by us and others for tissue concentrations of PQQ [see (1,7,8) and references cited therein].

Taken together, these observations suggest that compounds with the properties of PQQ may play an important role in oxidative metabolism. Although cofactor functions have been suggested and debated [(7); see also (52,53)], these results suggest that further investigation of the potential of PQQ potential to interact in redox-sensitive cell signaling pathway(s) or as an agonist or ligand important to mitochondriogenesis (e.g., peroxisome proliferator-activated receptor stimulation) may also prove to be fruitful (54,55).

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